AVIAN SOFT TISSUE SURGERY

David Sanchez-Migallon Guzman,
LV, MS, Dipl. ECZM (Avian, Small Mammal), Dipl. ACZM
Companion Exotic Animal Medicine and Surgery,
School of Veterinary Medicine University of California Davis
One Shields Avenue 2108 Tupper Hall Davis, California 95616
Email: guzman@ucdavis.edu

1. INSTRUMENTATION

Microsurgical instruments

Microsurgical instruments are preferred.[1] They should be a standard length (usually 6–7 inches long), have rounded handles and have small working tips. Many microsurgical instruments are also counter balanced and shaped to fit the notch between the base of the thumb and the index finger. When performing surgery, the hands rest firmly on the operating table. Both hands are positioned in the same manner as when we write, and the instruments are manipulated with the fingers only. The most important instruments in addition to the standard surgical instruments are the micro forceps, micro scissors, and micro needle holders. Other instruments can be added to the set, such as micro hemostats, vascular clamps, and others. DeBakey thumb forceps are atraumatic vascular forceps that can be used to handle delicate tissues.

Radiosurgery

Radiosurgery utilizes high frequency radiowaves to cut with little heat damage to the adjacent tissues.[2] The Surgitron Dual Frequency 120 (Ellman International, Inc., Hewlett, NY, USA) employs 4.0-MHz offers monopolar/bipolar applications with foot-pedal/finger switch control for cut and coagulate. The ground is a plate with an antenna and does not even have to be in contact with the patient. It has both bipolar and monopolar settings. The monopolar cutting needles are mostly used for dissection, while the bipolar forceps are used for sealing blood vessels. The Harrison bipolar tip have one arm of the tip is bent at a 45^{o} angle and are used for cutting avian skin. Radiosurgical units should not be used in the presence of flammable material.

Magnification and illumination

Magnification is recommended to handle small and delicate tissues in small birds.[3] The ideal surgical loupes system would be comfortable; provide magnification of 3–5X; have lenses with a large depth of field so more than one plane is in focus; have a focal distance appropriate to the surgeon's working distance, minimizing stresses on the neck and back (ergonomic); and allow the surgeon to look around the lenses to accomplish tasks that do not need magnification. Surgitel system (General Scientific Corporation, Inc. Ann Arbor, MI, USA) provides lenses that meet this criteria. Illumination with a focal light source is recommended. Lens-mounted lights direct the light where the surgeon needs keeping it in the field of view when the head is moved, but focal lights are also available separate from loupes.

Soft-tissue retractors

The soft-tissue retractors recommended are the smallest sized Balfour and Gelpi retractors for larger patients, and the Bennett avian retractor and the Doolen avian retractor (Sontec Instruments, Inc.) for smaller patients.[4] The Lone Star Retractor (Lone Star Medical Products, Inc., Stafford, TX, USA) is available in a variety of configurations but the rectangle, and it is likely the most versatile retractor used in avian surgery.

Vascular clips

Hemostatic clips, such as Hemoclips (Hemoclips and Samuels Hemoclip Applier, RICA Surgical Products, Inc.) are made of metal (steel, titanium, or tantalum) and serve the same purpose as ligatures.[4] They are available in various sizes. They are quick and easy to apply. It is recommended that the length of the clip be 2–3 times the diameter of the vessel. Right-angled appliers are also available and are especially useful for applications deep in the body cavity of small patients.

Sutures

Polydioxanone suture (PDS, Ethicon, Novartis, Somerville NJ, USA), polyglyconate (Maxon) and polyglecaprone 25 (Monocryl, Ethicon) are recommended absorbable sutures.[5,6] Polyglactin 910 (Vicryl, Ethicon) causes severe inflammatory response in birds, and it is only recommended for cloacopexy to promote adhesions in the surgical site[5,6] Polypropylene (Prolene, Ethicon), nylon and steel are recommended nonabsorbable sutures. The sizes 3-0 to 6-0 are commonly used in avian surgery. The fewest and finest sutures possible should be used, and efforts should be made to minimize tissue trauma to decrease the risk of would infection. Birds generally do not traumatize the suture lines, allowing the use of continues suture patterns in the skin.

2. PATIENT PREPARATION

The feathers should be plucked 2-3 cm from the surgical site to prepare the skin for surgery. Small feathers can be plucked three to four at a time, while contour and covert feathers should be plucked individually in the direction of their growth to avoid tearing or bruising the skin. Rectrices and remiges should be pulled one at a time in the direction of their growth or wrapped with bandage material if possible. Feathers that are plucked regrow quickly (usually nearly regrown in 2 weeks at suture removal); however, cut feathers do not grow back until their normal molt, which can be several months. The remaining of the feathers may be retracted from the surgical field with light adhesive tape such as masking tape.

The skin is prepared with routine cleansings with chlorhexidine alternating with sterile saline. The use of excessive amounts of water, alcohol, and chlorhexidine may result in hypothermia. Chlorhexidine is more commonly used because it has residual activity, has a broad spectrum of activity, causes fewer skin reactions, and is not inactivated by organic material. Saline is recommended instead of alcohol because has proven to be as effective in preventing infection when alternated with the disinfectant solution, and because alcohol use will result in a greater heat loss.

Clear plastic drapes (Veterinary Transparent Surgical Drape, Veterinary Specialty Products, Inc., Shawnee, KS, USA), allow the patient to be monitored more closely, for example, for respiratory rate and character. These drapes are placed in addition of the quarter drapes around the patient and the biggest drape to cover the table.

3. SURGERY OF THE SKIN

Digit constriction repair

Circumferential constriction caused by fibers, scabs, or necrotic tissue may result in avascular necrosis of the digit distal to the constriction. The fiber, scab, or constricting tissue should be remove to reestablish circulation to the digit, preventing circumferential scabs from forming following surgery by using a hydroactive dressing.

In neonates, proposed causes for necrosis of digits include low humidity. In the early stages, increasing the environmental humidity, hot moist compresses, and massage therapy may be effective. In more advanced cases, the indented tissue is excised and a circumferential skin anastomosis performed. Following the anastomosis a release incision should be made on the medial and lateral aspects of the digit longitudinally across the anastomosis, and a bandage with hydroactive dressing is applied.[7]

Feather cyst excision

Trauma or abnormal development of the follicle may result in the formation of a cyst. Canaries of the Norwich and Gloucester breeds are predisposed. Recurrence is seen following treatment by lancing and curettage. Blade excision of the affected follicle is the recommended treatment. Smaller cysts are easily removed using fusiform excision of a small piece of skin including the cystic feather follicle.[8] Some larger cysts may be saved by marsupializing the lining of the follicle cyst to the surrounding skin. An incision is made in the center of the cyst parallel to the direction of the feathers normal growth. The feather debris is removed and the redundant tissue is excised. A simple continuous pattern of a monofilament suture is used to appose the cyst lining to the skin. The growth of the new feather is closely monitored.

Cranial wound repair

Trauma to the cranium may result in a degloving wound. Small wounds may be allowed to heal by second intention under a hydrophilic dressing. Larger wounds might be close using a single pedicle advancement flap used to cover the defect on a bird's head, preventing stretching of the facial and upper eyelid skin that would have resulted from simple closure of the wound. Two skin incision at the lateral aspects of the caudal end of the defect are initiated to create the pedicle flap. Divergence of the lateral incisions creates a wider base that increases blood supply. The pedicle is undermined bluntly and advanced rostrally. Closure is routine.[9]

Sternal wound repair

Trauma to the keel may result in a splitting wound through the skin and deeper tissues, affecting even to muscle and bone. Small superficial wounds may be allowed to heal by second intention. Larger wounds might require surgical debridement of the affected skin, muscle and bones. The pectoral muscles are sutured together over the keel or anchored to the keel. Smaller defects can be closed by approaching the edges of the wound. Larger defects might be closed using a cutaneous axial flap. Two skin incisions on the side of the thorax are initiated. The skin is elevated to ensure that the caudal external thoracic artery and its associated vein were included in the flap at its base. The axial pattern flap is advanced forward with minimal tension to cover. The cranial edge of the flap is sutured to the cranial left corner of the defect. The caudal edge of the flap is sutured to the caudal left corner of the defect in a similar manner. Closure is routine.[10]

Uropygial gland excision

Adenoma, adenocarcinoma and squamous cell carcinoma are commonly described neoplasms of the uropygial gland in birds. The gland is excised initiating an incision around it. The gland has significant blood supply. The tissue is undermined starting caudally and extending circumferentially and cranially until the gland is removed while bleeding is controlled. Closure is routine. If the defect is too larger to allow primary closure, it may be allow to heal by second intention with hydroactive dressing.[11]

4. SURGICAL APPROACHES TO THE COELOMIC CAVITY

There are several surgical approaches to access to the gastro-intestinal tract, reproductive tract, lower respiratory tract, urinary tract, and other viscera. These include ventral midline, left or right lateral, transverse coeliotomy or combinations of the latest.[7] The patient is placed in dorsal recumbency for midline and transverse approach, or left or right recumbency for right and left approaches respectively. For any approach, the cranial part of the patient should be elevated 30-40 degrees to prevent fluids from flowing craniad and entering the lungs. Similarly, patients with ascites should have the fluid removed prior to coeliotomy. Because coeliotomy involves invasion of the air sacs, anesthetic gas may escape altering the ability to maintain anesthesia.

The left lateral coeliotomy access the air sacs to provide the best exposure to the proventriculus, ventriculus, and female reproductive tract. The left leg may be retracted caudally or cranially to increase exposure of cranial and caudal coelom respectively. The skin is incised over the left paralumbar region from the last rib to the caudal pubic bone. The superficial medial femoral artery over the lumber fossa should be cauterized. The abdominal muscles are incised. The last two ribs might be transected to provide adequate exposure to the cranial coelom.[7]

The ventral midline, transverse or combination coeliotomy provides best exposure of both sides of the coelomic cavity. The ventral mid-line approach made caudal to the ventriculus access directly the intestinal peritoneal cavity, and not the respiratory tract. For the midline approach, the skin is incised on the ventral midline from the caudal sternum to the interpubic space. The linea alba is identified, lifted and incised. Care must be taken to incise only the linea alba, because the duodenum lies directly inside the body wall. The incision can be extended transversally at the sternal border to allow greater exposure. For the transverse approach a skin incision is made midway between the sternum and the vent. The body wall is lifted and incised. Care must be taken to incise only the body wall. For both approaches, the body wall and skin are closed separately with 4-0 to 6-0 absorbable suture.[7]

5. SURGERY OF THE GASTROINTESTINAL SYSTEM

Ingluviotomy/crop biopsy

Crop impaction or foreign body may require removal via ingluviotomy. Chronic ingluvitis or a proventricular dilation disease suspect, may require crop biopsy. An incision is made through the skin over the left lateral portion of the crop. The skin is bluntly dissected to identify the crop. Stay sutures are placed in the crop wall. The crop is then incised at the cranial aspect of the left lateral side of the sac, and a tissue sample from a vascular region is collected if needed for diagnosis of proventricular dilation disease.[12] The crop is closed using 5/0 or 6/0 absorbable suture in a continuous suture pattern, with a second inverting continuous pattern if required to ensure complete closure. Skin closure is routine.[13] Provide small and frequent feedings post-operatively until the incision is healed.

Crop burn/laceration repair

Overheated formula hand-fed to neonates may result in crop burns with or without fistula formation. Trauma to the crop may result in lacerations. Surgery should be delayed 4-5 days in the case of a crop burn for declaration of the non-viable tissue. The affected tissue is resected, and closure is routine.

Proventriculotomy/Ventriculotomy

Removal of foreign or toxic material from the proventriculus or ventriculus may require proventriculotomy or ventriculotomy. The left lateral approach is used. The ventral suspensory structures must be dissected bluntly to retract the proventriculus caudally. Stay sutures are placed in the wall of the ventriculus to exteriorize both organs.The proventriculus is fragile and might tear if manipulated with toothed forceps or if stay sutures are placed. The coelomic cavity should be packed with moist gauze sponges. The proventriculotomy is initiated at the isthmus and extended cranially as needed. The ventriculus may be accessed through a proventriculotomy (preferred) or through a ventriculotomy. The foreign or toxic material is removed. The proventriculus is closed using a simple continuous pattern with a second inverting continuous pattern using a 4-0-6-0 absorbable suture. The ventriculus is closed with a simple interrupted pattern using a 3-0 to 5-0 absorbable suture.[13]

Enterotomy

Removal of foreign body, trauma and neoplasia may require enterotomy. An anastomosis is performed using a 6-0 to10-0 absorbable monofilament suture on a one-fourth-circle atraumatic needle. Typically six to eight simple interrupted sutures are necessary for end-to-end anastomosis. Abdominal wall and skin closure are routine.[13]

Ventplasty/Cloacopexy

Minor cloacal prolapse are reduced and held in place with simple interrupted or mattress sutures on both sides of the vent while the underlying cause is treated. Major or recurrent cloacal prolapse may require cloacopexy and/or ventplasty procedures. The underlying cause of the prolapse needs to be addressed to prevent recurrence. For cloacopexy a ventral midline coeliotomy is performed and the cloaca identified. The fat on the ventral surface of the cloaca must be excised. Circumcostal, sternal and subserosal pexy procedures may all be considered. The cloaca is sutured full thickness or partial thickness with incisions created in the serosa into the ventral midline body wall. For the first method, the suture passes through one side of the body wall, through the full thickness of the cloaca and through the other side of the body wall in a simple interrupted pattern. For the second, incisions are created in the serosal surface of both the body wall and the cloaca, paramedian approximately 5-10 mm from the midline. Three or four sutures are placed between each side of the 2 incisions such that the serosal surfaces are sutured and the subserosal surfaces are opposed.[14] For ventplasty, once the resection site is determined, excise the desired triangular area taking epidermis and dermis. When apposed, the new epidermal edges should form the desired vent diameter. First, close the submucosa with the dermis with simple interrupted suture. Next, close the dermis in the same fashion. Finally, the overlying epidermis is closed. The end result should be one suture line extending cranially (single vent resection) or one suture line extending laterally on the left and right sides of the vent (double vent resection). The new vent diameter should be just large enough to allow passage of droppings.

6. SURGERY OF THE RESPIRATORY SYSTEM

Rhinolith removal

Infection and malnutrition may result in rhinolith formation from desiccated secretions and debris. Mild cases may respond to nasal flushes with saline. Severe cases may require debridment. A stainless steel aural curette may be used to gently elevate and remove the mass from the naris. A lacrimal cannula may be used in smaller avian patients such as budgerigars and passerines. The nares should be flushed copiously after removal of the bulk of the mass to remove any small pieces or material and to assist in resolution of any pathogens. Samples should be collected for cytology and culture.[7]

Sinusotomy

Infection may result in periorbital swelling of the infraorbital sinus. Mild cases might respond to medical management. Severe cases may require sinusotomy. The skin is incised in the ventral aspect below or rostral to the eye, and the material accumulated in the infraorbital sinus is removed. The sinus is flushed copiously and the skin is closed with 4-0 to 5-0 absorbable suture or left open and allow to heal by secondary intention.

Tracheal resection and anastomosis

Trauma, infection or neoplasia may result in tracheal stenosis. The affected segment of the trachea may be resected. The surgery requires prior air sac intubation to maintain anesthesia. The skin is incised along the ventral midline of the neck or just lateral to the crop. The subcutaneous tissues are dissected to reveal the paired sternothyroideus muscles. The recurrent nerves are identified and avoided. A tracheal ring cranial and caudal to the affected segment is bisected circumferentially with a No. 11 scalpel blade. The tracheal ends are approximated by preplacing simple interrupted sutures of 4-0 to 6-0 PDS on a tapered needle. The sutures are tightened individually to appose the tracheal ends. Tension-relieving sutures are placed on the ventral and lateral aspects of the trachea, each encircling a tracheal cartilage proximal and distal to the anastomosis site. All suture knots are extraluminal. The sternohyoid muscles were apposed over the ventral aspect of the trachea. The skin closure is routine.[15]

Pneumonectomy

Infection or neoplasia may require partial pneumonectomy. The lungs may be approached through the caudal thoracic air sac or the dorsal intercostal spaces following retraction or removal of the ribs. The lung is elevated from the ribs and other structures using a spatula or moistened cotton tipped applicator. Hemostatic clips are placed on the lung tissue to isolate the portion to be removed. The portion distal to the clips is then removed. The skin is closed routinely.

7. SURGERY OF THE REPRODUCTIVE SYSTEM

Ovocentesis

If the egg cannot be removed by manual manipulation, ovocentesis may be required. To perform ovocentesis insert a large gauge needle (20-18 gauge) and aspirate the content of the egg while applying gentle finger pressure to collapse the shell. This procedure is most effective when the retained egg is thin shelled, as compared to a fully formed and rigid shell. The shell fragments are usually passed within a few days. If the egg is not visible through the vent, ovocentesis can be performed transcutaneously. Keep in mind that if significant oviductal pathology is present ovocentesis and collapsing the shell can lead to oviductal rupture. Severe cases of dystocia may need surgical treatment.[3]

Ovariectomy

Ovarian neoplasia, ovarian granulomas, persistent follicular activity, oophoritis and ovarian cysts that do not resolve with medical therapy may require ovariectomy. The ovary is tightly adhered to its dorsal attachments. This makes complete excision of the ovary difficult and poses significant risk of hemorrhage to the patient. The avian ovary is attached to the cranial renal artery by a short stalk and the attachment to the common iliac vein is intimate and extensive. There is a significant risk of damaging the left adrenal gland, significantly altering blood flow through the renal portal system and the cranial renal division, and damaging the overlying kidney and lumbar and/or sacral nerve plexus. Closure is routine.[14]

Salpingohysterectomy

Chronic egg laying, dystocia, neoplasia or infection unresponsive to medical therapy may require salpingohysterectomy. Although this procedure is described as a treatment for excessive egg laying, it is far from a complete address of the hypothalamic-pituitary-gonadal axis, and should be regarded as an incomplete treatment for the entire mechanism behind egg production. The left coelomic approach is preferred. The ventral ligament dissected to allow the oviduct to be released and positioned in a linear fashion. The oviduct can be removed moving from cranial to caudal, caudal to cranial, or in portions and segments as required. If starts cranially, elevate the fimbria of the infundibulum that lie caudal to the ovary to expose the dorsal ligament. A small blood vessel can be identified coursing from the ovary through the infundibulum and should be coagulated. The remainder of the suspensory ligament may then be dissected. With the infundibulum free, the oviduct is retracted ventrally and caudally exposing the dorsal suspensory ligament and the small vessels perpendicular to the oviduct and uterus. As dissection continues towards the cloaca, the ureter may be identified and must be avoided. The uterus courses along with the terminal colon and enters the cloaca. It is ligated at its junction with the vagina with suture or using 1-2 clips a short distance from the cloaca. Closure is routine.[14]

Cesarian

Breeding birds or unstable birds may require cesarean to preserve the integrity of the reproductive tract. The surgical approach varies with the location of the egg. If located cranially, a left lateral coeliotomy is recommended, and if caudally located, a ventral midline approach with or without a transverse incision provides optimal exposure. The oviduct is incised directly over the egg, avoiding obvious blood vessels, and the egg removed. The oviduct is closed with a simple interrupted or continuous pattern using an absorbable monofilament suture. Closure is routine.[14]

Orchidectomy

Testicular neoplasia, testicular cysts, and infectious and inflammatory conditions of the testicles that are unresponsive to medical therapy may require orchidectomy. The ventral midline approach or lateral approach is preferred. The testicle is gently retracted ventrally and a 90° vascular hemostat or hemostatic clip is applied to the mesorchium of one testicle with 90° clip applicators incorporating its vascular supply. The hemostat or clip must be applied parallel to the spine to avoid entrapping the aorta and peripheral nerves. If possible, a second clip is applied just ventral to the first. One clip is applied in a cranio-caudal direction and the second clip in a caudo-cranial direction to incorporate the entire vascular supply. The mesorchium of the testicle is incised between the hemostat or clip and the ventrally applied clips with a scalpel blade. Any remaining testicular tissue that protrudes through the hemostatic clip may be ablated with electrocautery or a laser. Residual testicular tissue may result in tissue hyperplasia and produce reproductive hormones. Closure is routine.[14]

Vasectomy

"Teaser males" and control of reproduction may require vasectomy. In the budgerigar, the patient is placed in dorsal recumbency and a 3- to 7-mm incision is made lateral to the cloacal sphincter. The fat and abdominal musculature is carefully dissected to enter the coelomic cavity. An operating microscope is used to locate the ductus deferens and a 5-mm section of the ductus deferens is excised. The skin is closed routinely.[16] In the finch, a 3-mm incision is made 5 mm lateral to the cloaca with the use of an operating microscope. The fat and abdominal musculature is incised to access the seminal glomera. The ductus deferens is carefully separated from the ureter and one or more sections excised without ligation. The skin is closed routinely.[17] Vasectomy in larger avian patients is performed via transection of the ductus deferens via lateral or transverse coeliotomy.[18]

Phallectomy

Chronic prolapse in male ducks may require amputation. The phallus is penetrated with absorbable 3-0-to 5-0 suture between the fossa ejaculatoria and the sulcus phalli, and sutured with a transfixion suture pattern. The phallus is cut distal to the ligature, and the stump is reintroduce in the cloaca.[19]

REFERENCES

1. Bennett R. Surgical Considerations In: Ritchie BW HG, Harrison LR, ed. *Avian Medicine: Principles and Application*. Lake Worth, FL: Wingers Publishing, 1994;1081-1095.
2. Altman R. Radiosurgery (Electrosurgery) In: Altman R,al. e, eds. *Avian Medicine and Surgery*. Philadelphia, PA: WB Saunders Co, 1997;767-772.
3. Bowles H, Odberg E, Harrison G, et al. Surgical resolution of soft tissue disorders In: Harrison GJ LT, ed. *Clinical Avian Medicine Volume II*. Palm Beach, FL: Spix Publishing, 2006;775-829.
4. Bennett RA. Cold knife or hot laser: instrumentation considerations. Proceedings of the Association of Avian Veterinarians Annual Conference 2010;221-236.
5. Bennett R, Yaeger M, Trapp A, et al. Histologic evaluation of the tissue reaction to five suture materials in the body wall of rock doves (Columbia livia). *J Avian Med Surg* 1997;11:175-182.
6. Pollock C, Wolf K, Wight-Carter M, et al. Comparison of suture material for cloacopexy. Proceedings of the Association of Avian Veterinarians Annual Conference 2006;31-32.
7. Bennett R, Harrison G. Soft Tissue Surgery In: Ritchie B, Harrison G,Harrison L, eds. *Avian Medicine: Principles and Application* Lake Worth, FL: Wingers Publishing, 1994;1096-1136.
8. Harrison G. Microsurgical procedure for feather cyst removal in a citron-crested cockatoo (Cacatura sulprurea citrinocristatala). *J Avian Med Surg* 2003;17:86-90.
9. Gentz E, Linn K. Use of a dorsal cervical single pedicle advancement flap in 3 birds with cranial skin defects. *J Avian Med Surg* 2000;14:31-36.
10. Ferrell ST, De Cock HE, Graham JE, et al. Assessment of a caudal external thoracic artery axial pattern flap for treatment of sternal cutaneous wounds in birds. *American journal of veterinary research* 2004;65:497-502.

11. Wise RD. Surgical removal of uropygial gland tumor in budgerigars. *Veterinary medicine, small animal clinician : VM, SAC* 1980;75:1601-1604.

12. Gregory CR, Latimer KS, Campagnoli RP, et al. Histologic evaluation of the crop for diagnosis of proventricular dilatation syndrome in psittacine birds. *Journal of veterinary diagnostic investigation : official publication of the American Association of Veterinary Laboratory Diagnosticians, Inc* 1996;8:76-80.

13. Forbes N. Avian gastrointestinal surgery. *Semin Avian Exotic Pet Med* 2002;11:196-207.

14. Echols M. Surgery of the avian reproductive tract. *Semin Avian Exotic Pet Med* 2002;11:177-195.

15. Guzman DS, Mitchell M, Hedlund CS, et al. Tracheal resection and anastomosis in a mallard duck (Anas platyrhynchos) with traumatic segmental tracheal collapse. *Journal of avian medicine and surgery* 2007;21:150-157.

16. Samour JH, Markham JA. Vasectomy in budgerigars (Melopsittacus undulatus). *The Veterinary record* 1987;120:115.

17. Birkhead TR, Pellatt JE. Vasectomy in small passerine birds. *The Veterinary record* 1989;125:646.

18. Samour J. Vasectomy in birds: a review. *Journal of avian medicine and surgery* 2010;24:169-173.

19. Lierz M. Phallus amputation in pet ducks. *Exotic DVM* 1999;1.3:51-54.